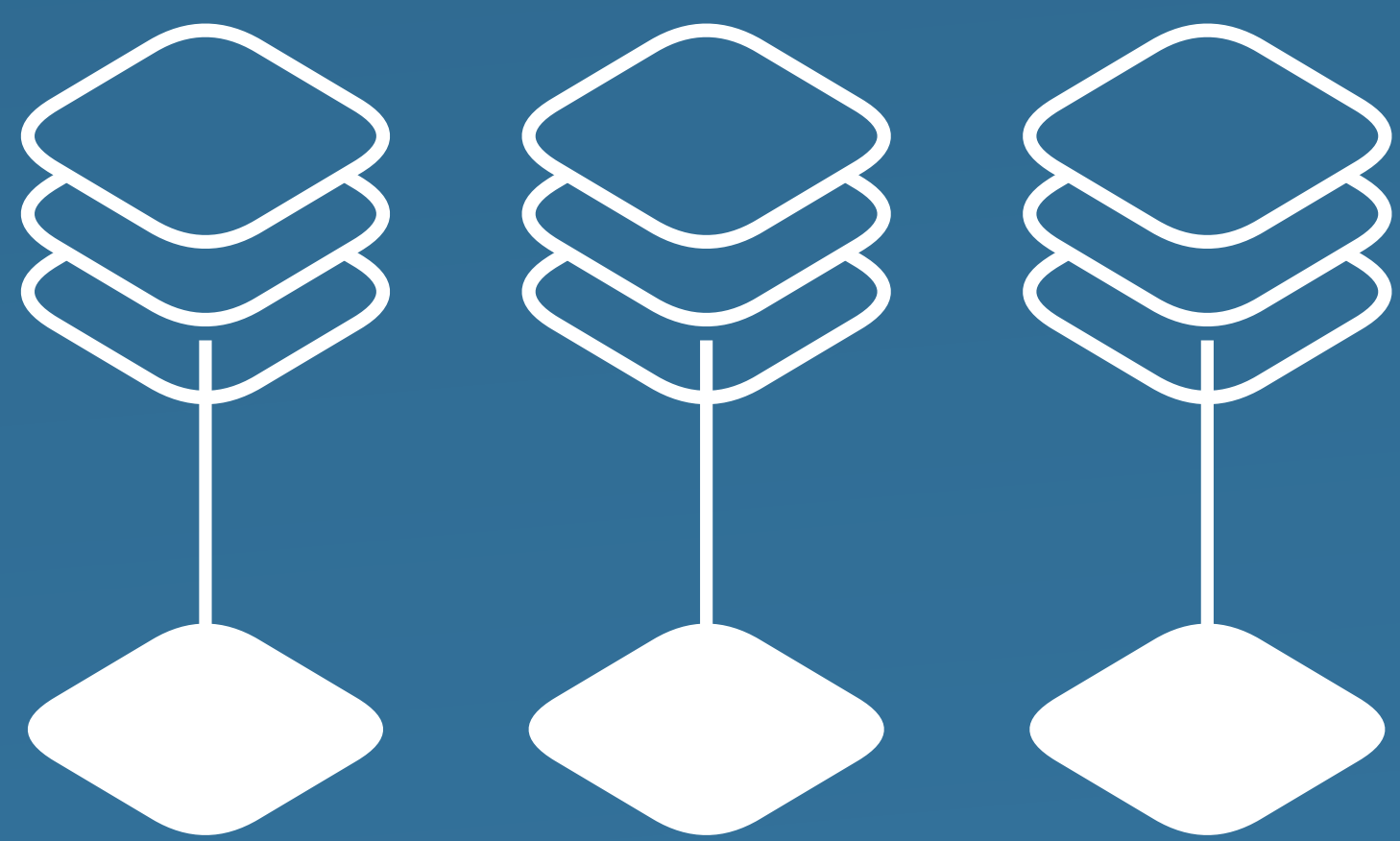
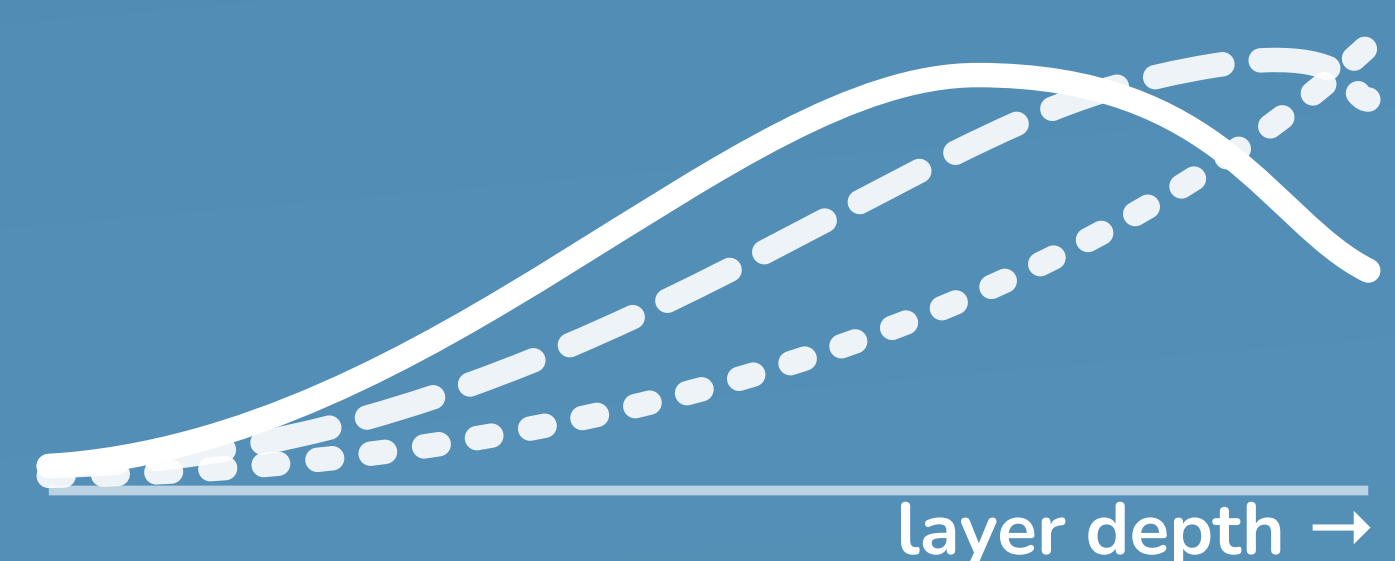




Which layer of a frozen vision foundation model should you probe?

None alone.

For dense prediction, try **AMDF**: it fuses **all** layers, for **every** patch.



On DINOv3-L, each NYUv2 task peaks at a different depth

≈22x

fewer params than concatenating layers (dense tasks)



Let's talk!

Open to research collaborations

vision foundation models · representation learning · probing & interpretability · spatial understanding



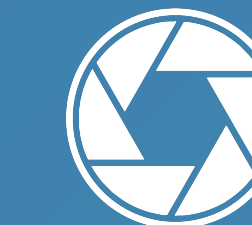
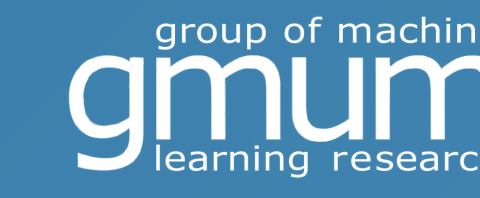
turhancankargin.me

AMDF: Holistic Probing of Vision Foundation Models with Adaptive Multi-Layer Dense Fusion

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Tasks live at different depths

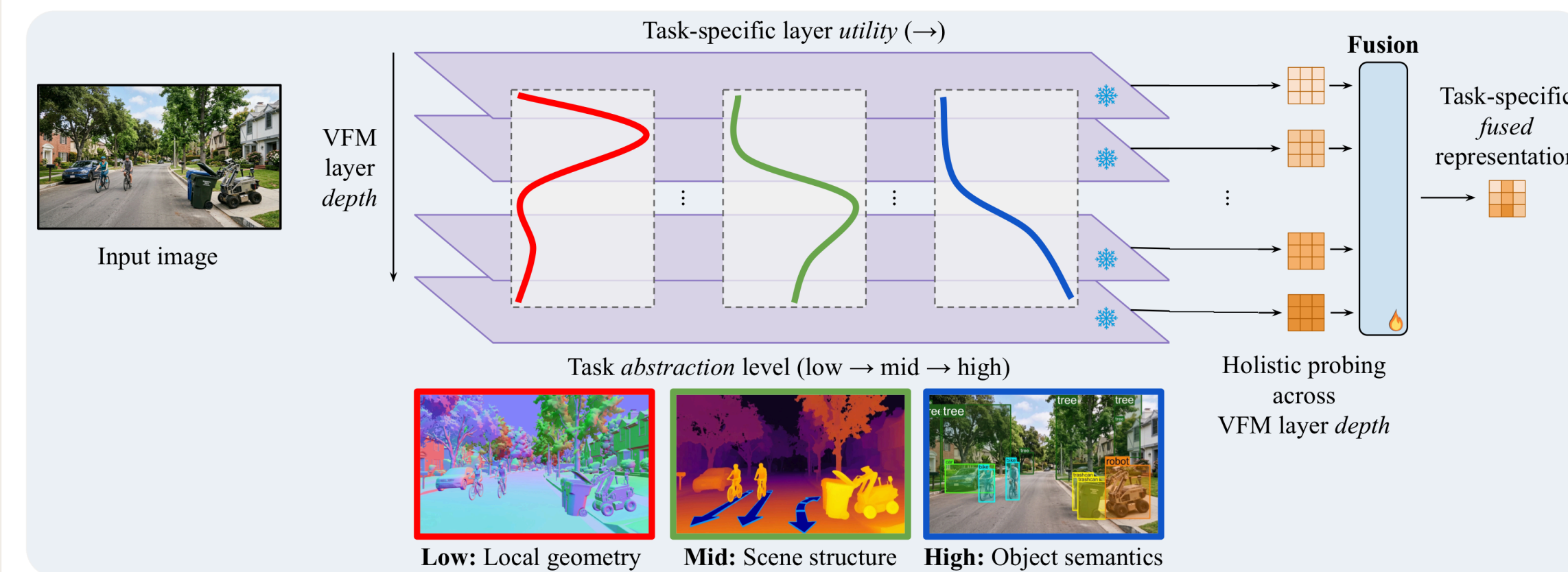


Fig. 1: Geometry peaks mid-depth, semantics at the end; AMDF fuses all layers per patch.

#1: last layer ≠ always best

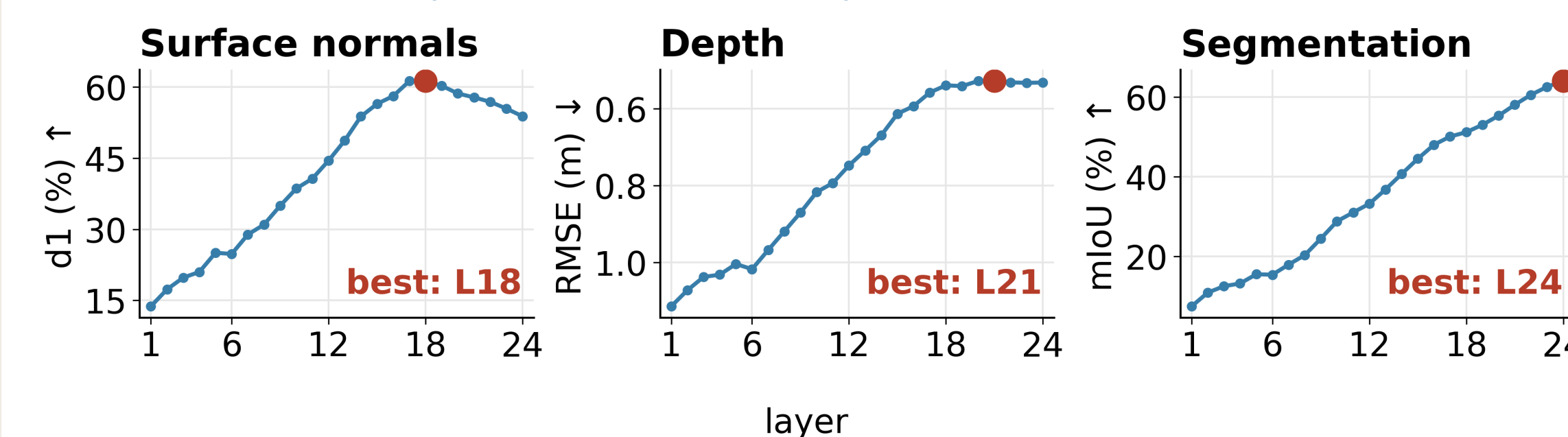


Fig. 2: Best single layer differs per task: ≈L18 (normals), ≈L21 (depth), L24 (segmentation); one seed per layer (DINOv3-L, NYUv2).

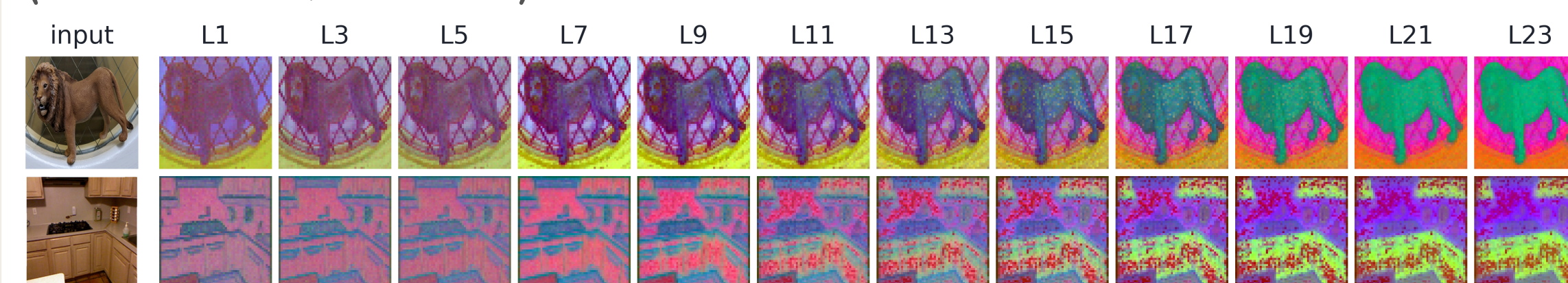


Fig. 3: Objects separate from the background only in deep layers (PESpatial-L, PCA).

#2: all layers = costly

✗ Concat all layers: **26.2M params**, the same fixed weights for every patch.

✓ AMDF: **per-patch** layer choice, **≈1.2M params**.

Take-home

- Useful features are **spread across depth**.
- AMDF: **2nd-best mean rank** with **≈22x fewer params** than Concat.
- AMDF beats every **fixed layer mix** by mean rank.

7 frozen VFMs

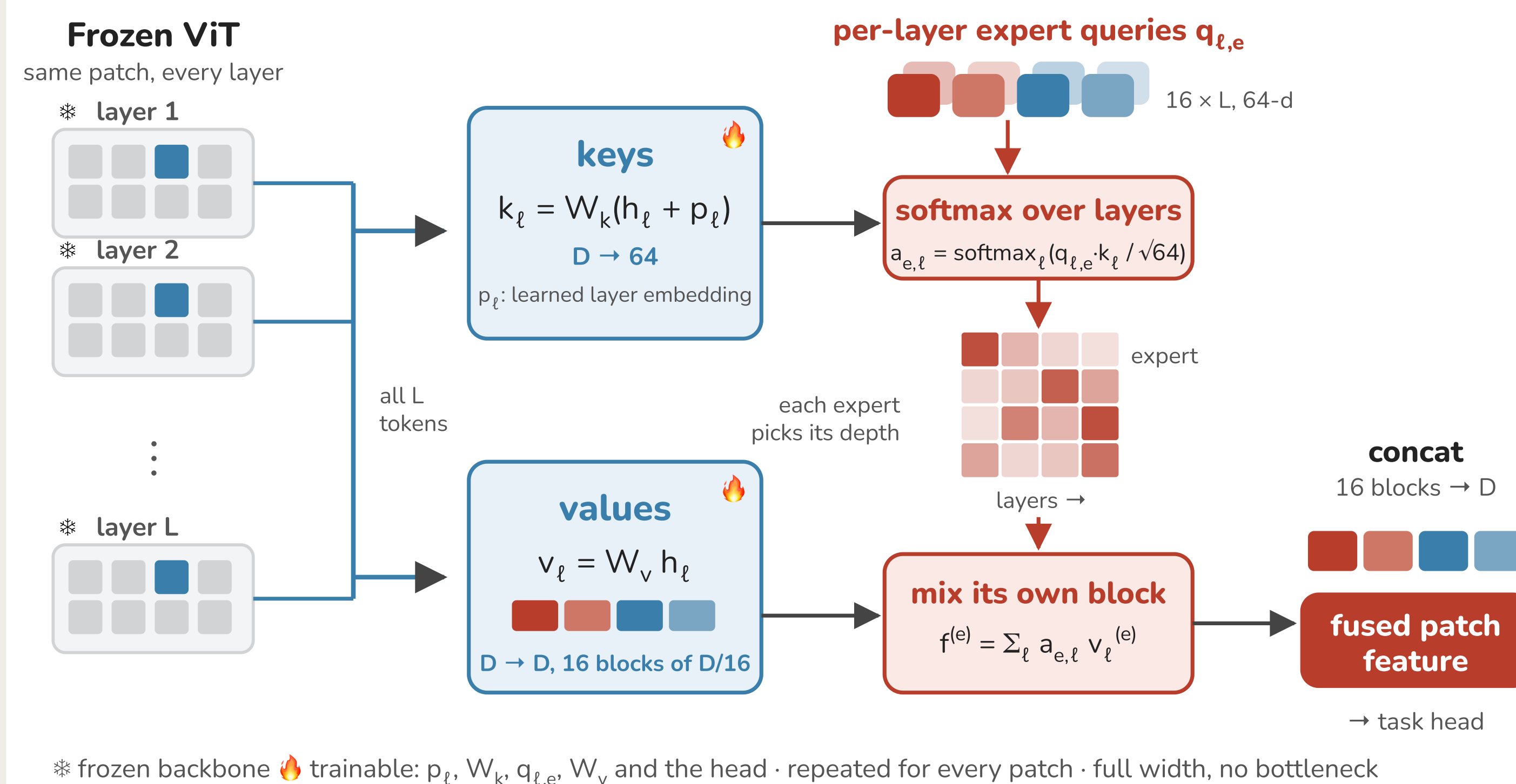
5 benchmarks

5 baselines

≤3 seeds

Solution: AMDF — depth-aware fusion

- Per patch, 16 experts with per-layer queries
- Softmax over layers → each expert picks its depth
- Frozen backbone, light task heads



✗ frozen backbone 🔥 trainable: $p_\ell, W_k, q_{\ell,e}, W_v$ and the head · repeated for every patch · full width, no bottleneck

Attention vs. layer utility (earlier variant)

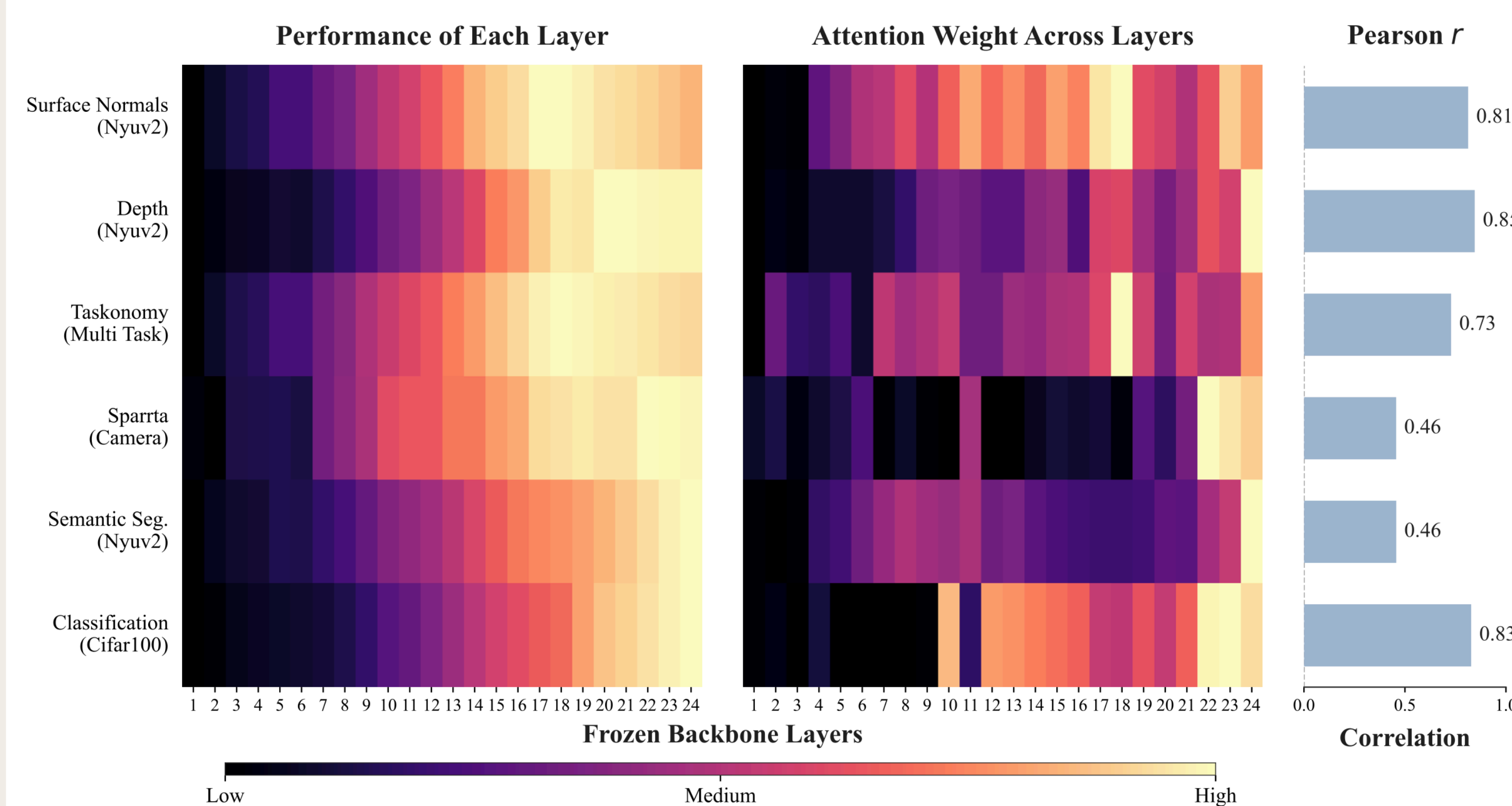


Fig. 4: Earlier AMDF variant (DINOv3-L): attention correlates with per-layer probe utility, $r = 0.46\text{--}0.85$ across 6 tasks.

Per-layer queries: better mean on all 5 tasks

Normals	Taskonomy	Segm.	Depth	Pose
+1.62 d1	-0.73	+0.16	+0.51 d1	-2.37° rot.
7/7 backbones	L1×100	mIoU	5/7 backbones	5/7 backbones
	7/7 backbones	6/7 backbones		

Paired gains vs. shared queries, mean over 7 backbones; cost: +2% params.

Results

≈22x fewer params than Concat
2nd mean rank (1st on depth)
5x7 benchmarks × backbones

Mean rank per benchmark

	Normals	Depth	Segm.	Taskonomy	Pose
AMDF (ours)	2.00	1.43 ★	1.86	2.00	1.93
ConcatLayers	1.00 ★	1.71	1.57 ★	1.00 ★	1.21 ★
WeightedLayers	4.14	4.00	4.14	3.29	3.89
SpatialWeights	3.14	4.14	4.43	4.00	4.00
LastLayer	5.43	4.14	3.86	5.29	4.96
AvgLayer	5.29	5.57	5.14	5.43	5.00

Fig. 5: Mean rank per benchmark (★ best). Concat 1st on 4/5, AMDF 1st on depth.

Accuracy vs. parameters

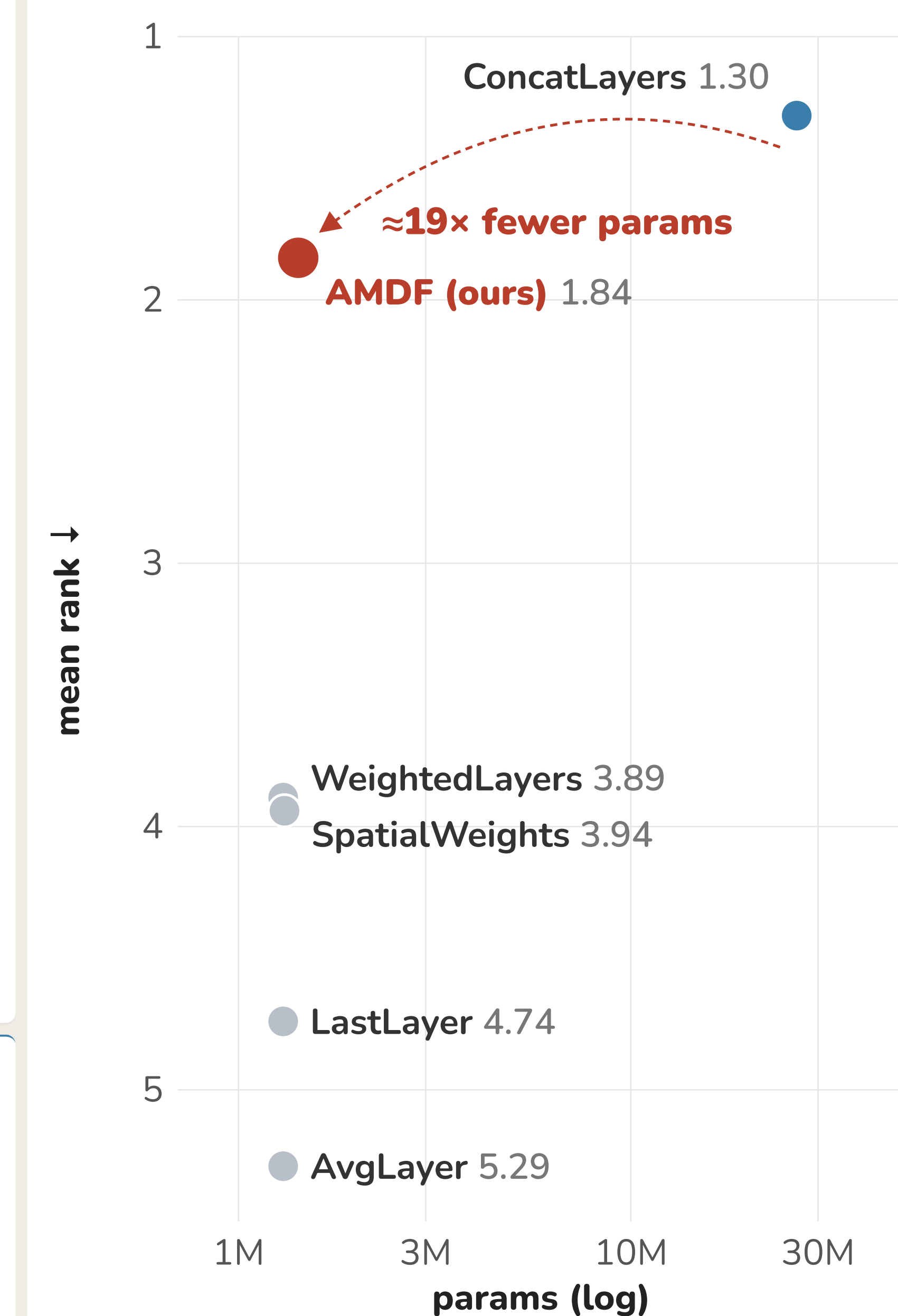


Fig. 6: ≈19x fewer params than Concat (5-task average, DINOv3-L).